DevUP backend team handoff

Post clone implementation and release guide

This handoff covers the DevUP repository from the November 2, 2026 clone through WU-21. It brings together shipped behavior, API contracts, schema changes, and the checks the backend team needs before deployment.

Scope and provenance

Clone baseline `1d1e7d2`. Upstream integration checkpoint `ec8beaf`. Handoff target `0cdf054` on `frontend`. The implementation is organized below by product area and ticket. The upstream GitHub, logout, file-deletion and lint changes are identified separately.

Deployment sequence

Back up the target MySQL database and inspect its migration history. Apply the pending migrations listed in the schema section, deploy the backend, then deploy the frontend. The older backend rejects some newer query parameters and does not understand note revision writes. Preserve uploaded file storage and the GitHub encryption key across deployments.

Verification baseline

The WU-21 frontend and backend builds passed, along with 607 backend tests. Local MySQL and Socket.IO checks exercised key workspace flows. Browser checks covered Files previews and Notes revision restore. GitHub remote outcomes and mail delivery were simulated in the relevant integration checks.

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Workspace foundation and task workflows

Upstream integrations and workspace foundation

The post-clone upstream updates added token-version logout with chat disconnection and GitHub OAuth with encrypted repository credentials, manual sync, commits and branch retrieval. The current GitHub integration supports a single eligible public repository per workspace. Upstream also added file-deletion retries and backend lint cleanup. The DevUP branch built the responsive landing page and authenticated workspace, then connected Home, Backlog, Board, GitHub, Activity, Team, Chat, Files, Notes, Analytics and Settings to backend services.

Task creation and Home

WU-01 prevents duplicate submissions and surfaces validation, API failures and retries. WU-07 prioritizes My work and Needs attention, adds task-list drill-downs from Home metrics, and retains workspace-wide progress. The workspace shell added search, notifications, task drawers, collapsible navigation and shared metric cards. The sidebar tooltip overflow fix is `3468c63`.

Deadlines and saved backlog views

WU-06 accepts `dueDate` as `YYYY-MM-DD` or `null` and keeps legacy `dueAt` for compatibility. Next 7 days includes today and the following six local dates. Done and cancelled tasks are excluded from due-date filters. WU-08 adds scope, sorting, search and filters. Named views use `GET/POST /workspaces/:id/backlog-views` and are isolated per workspace member. The current filter state is remembered in the browser.

Board columns placement and task identity

WU-09 persists column IDs, order, labels, limits and exact card placement. `POST /tasks/:id/transition` accepts `boardColumnId`, `placement` and `anchorTaskId` for a board move. Individual moves can target any visible column and adopt its workflow status. The original workflow rules still apply to status-only calls. Owner-only `PATCH /workspaces/:id/kanban-settings` manages the board, including safe deletion with a destination for affected tasks. Reopening a card clears its terminal timestamp. WU-10 adds stable `TASK-123` IDs alongside UUID links, quick-create assignee, priority and deadline, recoverable field edits, copyable links and one-step reassignment.

Onboarding invitations and workspace settings

WU 14 First run and account recovery

Signup opens a skippable avatar picker with five presets, HTTPS image URLs and local JPEG, PNG or WebP uploads. Uploads are converted to a small JPEG and saved through `PATCH /users/:id` as `avatarImage`. The avatar field can store both existing URLs and uploaded data. Email verification has a separate screen and preserves the intended destination. `POST /auth/email-verification/request` and `/confirm` handle verification. `POST /auth/password-reset/request` and `/confirm` handle reset. Links are single use. Verification lasts 24 hours and reset lasts 30 minutes.

WU 15 Invitations and team management

Workspace owners can invite an email address before signup, review status, resend or revoke. Links expire after seven days. Resend rotates the token. Acceptance requires the invited email address, and invite context survives signup and login. If delivery is unavailable, the owner can share the generated link. Existing in-app invitations still work. The API adds invitation `redeem` and `resend` actions, `DELETE /workspaces/:id/membership` and `PATCH /workspaces/:id/owner/:userId`.

Membership and authorization

Owners can transfer ownership to an existing member and remain a member afterward. An owner must transfer ownership before leaving. Leaving or removal unassigns the affected user’s tasks within that workspace and removes their workspace chat and unread subscriptions. Authored content and activity remain. Admin and guest permissions are documented as proposals and are pending implementation.

WU 16 Workspace settings and archive behavior

Search appears at 11 entries. The full picker counts active and archived workspaces. The sidebar switcher counts active workspaces. Recent history is per account in the browser, and switching accounts opens sign-in. Profile, avatar, security, theme and notification badge are personal settings. Access, board rules, GitHub, reporting time zone and archiving are workspace settings. The default reporting zone is `Africa/Casablanca`. Archived direct links show a banner. Shared writes are blocked and authorized reads, owner restoration, invitation revocation or decline, leaving and personal actions remain available.

Chat files and notes

WU 03 Chat unread state

Workspace and direct messages use REST for history and sending and Socket.IO for live delivery. Workspace read cursors persist per membership and move forward only, including when messages share a timestamp. `GET /workspaces/:id/messages/unread-count` and `PATCH /workspaces/:id/messages/:messageId/read` serve workspace read state. `GET /direct-conversations/unread-count` counts unread direct messages across conversations. The sidebar combines counts. Voice and video buttons remain visible and report that calling is unavailable.

WU 02 and WU 19 File library

Workspace file and note lists support server-side search, sorting and pagination. `lookahead=1` returns an extra row for the Next control. Files filter by `kind` and `uploaderId`. Uploads display progress and enforce a 10 MiB maximum. File details link to an associated task. Authorized members can download, and the uploader or owner can rename via `PATCH /workspaces/:id/files/:fileId`. The preview drawer renders supported raster images and plain text or CSV up to 256 KiB. Other formats use download. Attachments use download disposition and `nosniff`.

WU 19 Note reading and revision history

The split-pane Notes page provides title and content search, shared pins, author and updated metadata, deep links and references to workspace tasks by readable ID. Members can read notes and history. The creator or workspace owner can edit, pin, delete and restore. Archived workspaces allow reading. Direct links fetch notes by ID independently of the list page.

Note write compatibility

`PATCH /workspaces/:id/notes/:noteId` requires the current integer `revision`. A stale revision returns HTTP 409, and the frontend preserves the draft for review. The `revisions` routes list and preview earlier snapshots. `POST /workspaces/:id/notes/:noteId/revisions/:revision/restore` takes the current revision in its request body. Restore saves the version it replaces and retains the pin. Apply migration `1795200000000` and release both app layers together.

Projects workspace history and analytics

WU 17 Projects and milestones

`/workspaces/:id/projects` manages projects, milestones and task links. Projects have an optional owner and target date plus a planning status. Each task may link to one project and one optional milestone in that project. Existing tasks can stay ungrouped. Task details and projects link in both directions. Progress counts Done over linked non-cancelled tasks, independently of workspace progress or the project's planning status. Deleting a milestone keeps its tasks in the project. Deleting the project unlinks its tasks. Removing the project owner from the workspace clears ownership.

WU 04 and WU 20 Activity and audit

`GET /workspaces/:id/activities/page` filters by category, actor and date before pagination. The six categories cover tasks, files, notes, members, board settings and GitHub. New events include assignment, unassignment, blockers, file renames, note changes, board settings and GitHub sync outcomes. Task, file and note links show Deleted when the target is gone. Owner-only `GET /workspaces/:id/activities/audit` records workspace access changes, invitations, ownership transfer and archive actions. Legacy membership removals are audit-only. Ownership transfers and note edits record history in their transaction. Other event writes are best effort.

WU 21 Status history analytics

Current open, overdue and active counts are independent of the report range. Active means In progress or Review. `GET /workspaces/:id/analytics?days=7|30` uses the workspace time zone and status history for daily completion events, stage time and cumulative flow. Charts and tables share the report. Flow drill-downs use the same `asOf` timestamp and a selected `day`. End-of-day flow uses the last known status. Today uses the report time.

Historical reporting rules

Reopened tasks can contribute a later completion when they return to Done. Duration sums elapsed time in In progress and Review that overlaps the report range. Cancelled tasks retain prior stage time and appear in the Cancelled bucket. Deleted tasks leave flow at deletion and retain earlier history. Older tasks with incomplete history use available status and legacy timestamps. Current snapshot counts exclude cancelled and deleted tasks.

Database migration checklist

Apply only pending migrations in timestamp order. Compare the target database with TypeORM migration history first. The initial schema `1789763235441` is already part of the clone baseline.

Migration	Schema and rollout effect
1793491200000	Upstream token version. Allows logout to invalidate issued sessions.
1793900000000	Upstream encrypted repository OAuth credential storage. Preserve the encryption key.
1794300000000	Custom board layout and task column ID. Null values fall back to workflow defaults.
1794400000000	Workspace chat read cursors on membership records.
1794500000000	Calendar deadlines. Backfills legacy timestamps in Africa/Casablanca and retains the original values.
1794600000000	Named backlog views stored on workspace memberships.
1794700000000	Unique, stable task numbers for readable TASK identifiers. UUIDs remain valid.
1794800000000	Account verification and password reset token digests and expiry fields.
1794900000000	Expands avatar URL storage to TEXT for compressed avatar uploads.
1795000000000	Workspace reporting time zone and per-user theme and notification badge.
1795100000000	Projects, milestones and task links with workspace and membership relationships.
1795200000000	Note revision number, pin timestamp and revision snapshot table.

Rollback planning

Use a database backup for rollback planning. Down migrations can remove feature data, including custom column layouts, saved views, uploaded avatar data, projects and revision history. WU-20 and WU-21 reuse the existing activity and task-history tables.

Deployment checks and remaining work

Configuration and deployment

Use the repository's Docker Compose migration service or the backend `migration:run` script. Check migration status before release. Route `/api/*` to Nest with the frontend prefix stripped and forward `/socket.io` with polling and WebSocket upgrade. Preserve MySQL and the file storage volume. Deploy matching frontend and backend builds.

Email and GitHub verification

Configure `AUTH_PUBLIC_URL`, `AUTH_EMAIL_FROM` with a verified sender, and `RESEND_API_KEY`. Test signup verification, password reset and invitations with real inboxes. GitHub requires `GITHUB_OAUTH_CLIENT_ID`, `GITHUB_OAUTH_CLIENT_SECRET`, `GITHUB_OAUTH_CALLBACK_URL`, `GITHUB_OAUTH_FRONTEND_URL` and `GITHUB_TOKEN_ENCRYPTION_KEY`. Register the exact callback. Verify a live GitHub authorization and sync, revoked access and recoverable errors.

Release acceptance checks

Use an owner, member and outsider. Verify workspace isolation, invite expiry and replay, chat room removal, archive guards and audit ownership transfer. Check board placement after reload, file preview and download permissions, concurrent note edits and restore, project links, and Analytics totals against known task history. Repeat realtime scenarios in two browsers for staging sign-off.

Backlog still open

WU-11 to WU-13 are partially covered by existing attachments, task history and command search. Bulk actions, checklists, subtasks, labels and the full command and deep-link scope remain open. WU-18 remains open for GitHub reconnect and disconnect improvements, PR-to-task context, signed and deduplicated webhooks and opt-in transitions. Private and multiple repositories are separate scope. WU-22 to WU-25 remain open for richer notifications, chat reliability, collaboration and later extensions. Calling, reactions, mentions and conversation-authorized attachments require more backend work.

Repository references

`docs/development/WORKSPACE_IMPROVEMENT_BACKLOG.md` contains the ticket-level scope and verification notes. `docs/development/WORKSPACE_BACKEND_HANDOFF.md` records the earlier integration checkpoint. Migration source is under `apps/backend/src/database/migrations`. The release target for this handoff is `0cdf054`.